

SOFTWARE REQUIREMENTS SPECIFICATION

SNAKE GAME

SRS Documentation

Project: Snake Game

Document Version: 1.0

Prepared for College Project Submission

1. Introduction

The Snake Game is a simple interactive game in which the player controls a snake. The snake moves around the game board and eats food to increase its length and score. The game ends when the snake collides with the wall or its own body.

2. Purpose

The purpose of this project is to develop an easy-to-use Snake Game that:

- Allows the player to control the snake.
- Provides food for the snake.
- Increases the score when food is eaten.
- Detects collisions.
- Displays Game Over when the snake collides.
- Allows the player to restart or exit the game.

3. Scope

The system provides:

- Start Game
- Snake movement
- Food generation
- Score calculation
- Collision detection
- Pause and Resume
- Game Over
- Restart Game
- Exit Game

4. Overall Description

4.1 User

The main user of the system is the Player.

4.2 Product Description

The player interacts with the game using the keyboard. The game system processes the input and updates the snake, food, score, and game status.

5. Functional Requirements

ID	Requirement
FR1	Player can start the game.
FR2	Player can move the snake Up, Down, Left and Right.
FR3	System generates food on the game board.
FR4	System detects when the snake eats food.
FR5	Score increases when food is eaten.
FR6	Snake grows after eating food.
FR7	System detects collision with wall or snake body.
FR8	Game ends when collision occurs.
FR9	Player can pause and resume the game.
FR10	Player can restart the game.
FR11	Player can exit the game.

6. Non-Functional Requirements

- **Usability:** Game should be simple and easy to understand.
- **Performance:** Snake movement and keyboard input should respond quickly.

- **Reliability:** Score and game state should work correctly.
- **Maintainability:** Code should be organized and easy to modify.
- **Portability:** Game should run in the selected development environment.

7. Hardware Requirements

- Computer/Laptop
- Keyboard
- Monitor/Display
- Basic RAM and storage required for the selected software

8. Software Requirements

- Operating System: Windows/Linux/macOS
- Programming language: Java/C++/Python or the language selected for implementation
- IDE/Code Editor
- Required compiler/interpreter

9. User Interface Requirements

The game interface should contain:

- Game board
- Snake
- Food
- Score
- Start option
- Pause/Resume option
- Restart option
- Exit option
- Game Over message

10. System Workflow

Start → Initialize Game → Display Board → Move Snake → Check Collision → Check Food → Update Score → Continue Game → Game Over → Restart/Exit

11. Use Case Description

Actor

Player

Main Use Cases

- Start Game
- Control Snake

- Eat Food
- Increase Score
- Pause Game
- Resume Game
- Game Over
- Restart Game
- Exit Game

12. UML Diagrams

The SRS can contain the following UML diagrams:

- **Use Case Diagram:** Shows the interaction between Player and Snake Game.
- **Activity Diagram:** Shows the step-by-step game process.
- **Sequence Diagram:** Shows interaction between Player, Game UI and Game Logic.
- **Class Diagram:** Shows classes such as Game, Snake, Food, Score and Board.
- **State Diagram:** Shows states such as Menu, Playing, Paused and Game Over.
- **Component Diagram:** Shows the major software components and their relationships.

13. Future Enhancements

- Different difficulty levels
- High-score storage
- Sound effects
- Background music
- Different game themes
- Multiplayer mode
- Mobile version

14. Conclusion

The Snake Game SRS document defines the purpose, scope, functional requirements, non-functional requirements, hardware/software requirements, system features and UML diagrams of the project. It acts as a guide for developing and testing the Snake Game.